Additional file 1

Specification records, numerical evidence and analysis code

Integrative pan-cancer and PAAD-focused multi-omics analysis of mucin-type O-glycosylation enzyme transcriptional landscapes

This supplement contains dated specifications and amendments, six-system comparison tables, Figure 1-4 and Figure 7 numerical tables, the frozen GIScore model, archived primary scripts, freeze manifests and a Figure 6E audit reconstruction. Large source objects and pretrained UNI2-h features are referenced in place rather than redistributed.

Six-system comparison in PAAD

System	n	HR C2/C1	Log-rank P	Global BH
NEDD8	177	1.020	0.9322	0.971
SUMO	177	1.561	0.04153	0.2492
OST	177	1.395	0.1113	0.4591
GALNT	177	2.253	0.000834	0.01376
GPI	177	1.401	0.119	0.4714
UFM1	177	1.558	0.03669	0.227

The system choice followed the six-system results. PAAD was retained below the earlier n >= 200 eligibility line. Historical specification documents are included verbatim; the manuscript and this audit explicitly qualify later deviations and outdated functional shorthand.

Specification and model chronology

The original Figure 4 model record gives a freeze timestamp of 2026-10-12 13:34:07. This is an archival record, not an independently timestamped registration. The model was selected on TCGA-PAAD ductal cases (n = 146); GEO validation used the archived coefficients. The full-precision coefficients are extracted directly from FROZEN_model.rds and the paper displays them rounded to three decimals.

GIScore = -0.344 z(GALNT16) + 0.202 z(GALNT14). External preprocessing uses within-cohort gene standardisation and a cohort median threshold. Consequently the coefficients are fixed, whereas the centring, scaling and group threshold are not a transferable locked clinical assay.

The original model SHA-256 is 67f48c019b747f7deefe261b61de7dcf5094414023dcf1247341de9e6f9fa8f4.

The code folder supplies original archived scripts with original checksums and the numbered pipeline. Some scripts contain historical machine paths, depend on existing score caches or processed single-cell objects, or write into original result folders. The code path inventory makes these dependencies explicit. A full pipeline execution has not been performed during packaging; work should use a separate configured workspace.

Source access

Data sources and public identifiers are listed in provenance/data_sources.csv. Elhossiny epithelial pseudobulk is the deposited pseudobulk_epi.rds: its local MD5 matches the v3 Zenodo record. UNI2-h features require institutional access approval and cannot be redistributed under the provider's access terms.

Figure 6 provenance and audit

The original Figure 6 preparation, primary-analysis and publication scripts (31/32/33) are present. The missing archival component is the writer of H1_panin_enzymes_robustness.csv, used by panel E. Its numerical contents can be reproduced independently; the replacement audit script is explicitly labelled as a reconstruction and is not represented as the recovered historical command.

The archived table is reproduced using 608 retained epithelial spots on seven slides. Primary model: log1p(counts/library x 10,000) ~ grade + (1|slide). Adjusted model: expression ~ grade + acini% + collagen% + log10(library UMI) + (1|slide). Grade is normal=0, low=1, high=2. Models use REML and two-sided Satterthwaite t tests. Categorical contrasts use grade as a factor. Detection fractions and slide direction counts use the same retained spots.

All 84 numeric fields across six rows match the frozen CSV within relative tolerance 1e-10. The GALNT4 adjusted fit is singular; this is retained and disclosed rather than silently replaced. 'adj. p' means a covariate-adjusted model P value, not BH adjustment. The primary all-gene screen retains its original BH family.

To repeat the audit from the attachment, unpack Additional file 4; set PROJECT_ROOT to a writable workspace containing its audit_input file at 02_intermediate/fig6_visium/spots_galnt_annot.csv.gz and the archived robustness CSV at its original 04_results/Figure6 path. Run verify_figure6.R. The script writes audit outputs to a dated report folder and preserves source evidence.

Reproducibility scope

Frozen figures, result tables and GIScore are unchanged. Upload archives contain exact copies with per-file SHA-256 records. The workbook carries full source precision; formatted display precision does not alter the stored values. Numerical Figure 6 reproducibility was tested. Other full pipelines, GIScore training, H&E fitting and upstream single-cell annotation were not rerun.

The original analysis sessionInfo / pip-freeze records were not present in the inspected primary package. The supplied R sessionInfo records the Figure 6 audit environment; packaging_environment.json records the current packaging tools. Neither is described as the original whole-study environment.

Known limits are preserved: the internal model-selection CV is not a separately nested evaluation of pipeline selection; external ABMIL model construction precedes its loop seed; some source annotations and caches originate outside this project. The supplementary figure SVGs are unchanged frozen files. Figure S2A retains its previously documented crowded labels and requires a separate approved layout revision for final upload.

The additional-file archive supplies code and supporting derived tables directly; no project repository deposit is claimed. Full raw matrices, large AnnData/Seurat objects and gated pretrained embeddings remain available through their source archives and documented local input contracts.